2520030117
skill-05

EXPERIMENT – 5

Single and Double Quote Handling in Custom Linux Shell

Aim:

To implement single and double quotes in a custom Linux shell, preserve quoted content and spaces, handle variable expansion, and test edge cases.

Objective:

- To understand single quotes.
- To understand double quotes.
- To preserve spaces inside quotes.
- To handle variable expansion.
- To detect unmatched quotes.
- To test quoted commands.

Theory:

A shell reads commands from the user and executes them.

Single quotes (`' '`) treat the content as literal text. Variables are not

expanded. Example:

echo '$HOME'

Output:

$HOME

Double quotes (`" "`) preserve spaces and allow variable expansion.

Example:

echo "$HOME"

Output:

/home/kruthi

Algorithm:

1. Start the shell.
2. Display the prompt `2520030340_kruthi>`.

3. Read the command.
4. Check for single and double quotes.
5. Preserve the content inside quotes.
6. Do not expand variables inside single quotes.
7. Expand variables inside double quotes.
8. Check for unmatched quotes.
9. Execute the command using `fork()` and `execvp()`.
10. Repeat until `exit` is entered.
11. Stop.

Functions Used:

Function	Purpose
`printf( )`	Display prompt
`fgets()`	Read command
`fork()`	Create process
`execvp( )`	Execute command
`wait()`	Wait for process
`getenv( )`	value Store
`strdup( )`	tokens
Get variable	

Sample Input:

2520030340_kruthi> echo 'hello world'
2520030340_kruthi> echo '$HOME'
2520030340_kruthi> echo "$HOME"

Sample Output:

hello world
$HOME
/home/kruthi

Observation:

1. Single quotes preserved literal content.
2. Spaces inside quotes were preserved.
3. Variables were not expanded inside single quotes.
4. Variables were expanded inside double quotes.

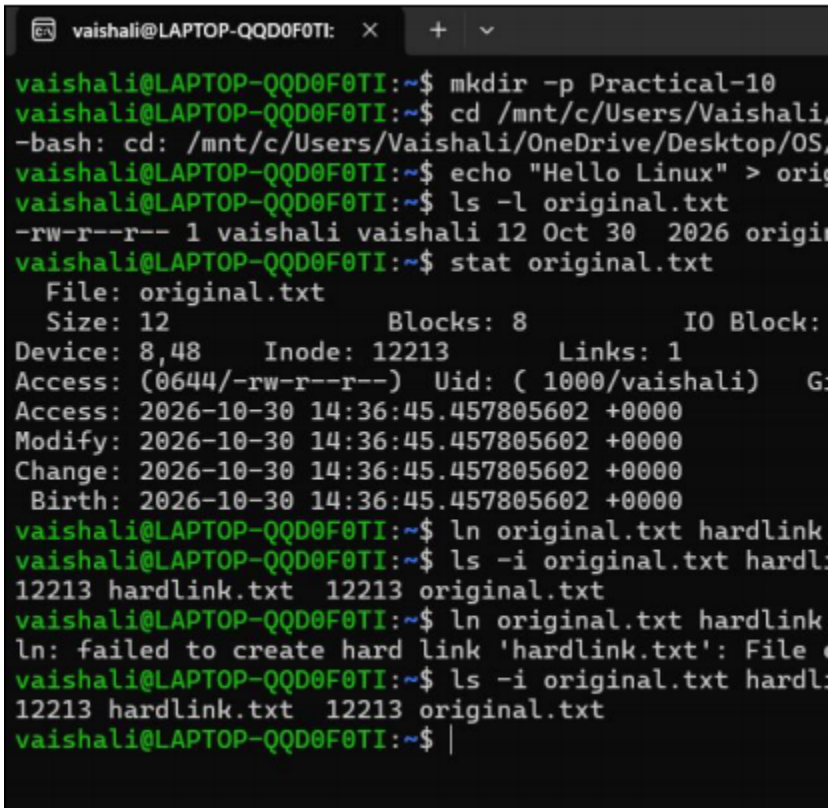
5. Unmatched quotes were detected.

Result:

The custom shell successfully handled single quotes, double quotes, spaces, variable expansion, and unmatched quotes.

Conclusion:

The experiment successfully implemented **quote handling in a custom Linux shell** and tested different quoted commands and edge cases.



```
vaishali@LAPTOP-QQD0F0TI:~$ ln -s original.txt symlink.txt
ln: failed to create symbolic link 'symlink.txt': File exists
vaishali@LAPTOP-QQD0F0TI:~$ ln original.txt hardlink.txt
ln: failed to create hard link 'hardlink.txt': File exists
vaishali@LAPTOP-QQD0F0TI:~$ stat original.txt
  File: original.txt
  Size: 12              Blocks: 8          IO Block: 4096   regular file
Device: 8,48    Inode: 12213       Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/vaishali)   Gid: ( 1000/vaishali)
Access: 2026-10-30 14:36:45.457805602 +0000
Modify: 2026-10-30 14:36:45.457805602 +0000
Change: 2026-10-30 14:37:05.452444785 +0000
 Birth: 2026-10-30 14:36:45.457805602 +0000
vaishali@LAPTOP-QQD0F0TI:~$ stat symlink.txt
  File: symlink.txt -> original.txt
  Size: 12              Blocks: 0          IO Block: 4096   symbolic link
Device: 8,48    Inode: 12242       Links: 1
Access: (0777/lrwxrwxrwx)  Uid: ( 1000/vaishali)   Gid: ( 1000/vaishali)
Access: 2026-10-30 14:38:01.696877251 +0000
Modify: 2026-10-30 14:37:57.009792289 +0000
Change: 2026-10-30 14:37:57.009792289 +0000
 Birth: 2026-10-30 14:37:57.009792289 +0000
vaishali@LAPTOP-QQD0F0TI:~$ find . -name "original.txt" -o -name "hardlink.txt" -o -name "symlink.txt"
./symlink.txt
./hardlink.txt
./original.txt
vaishali@LAPTOP-QQD0F0TI:~$ echo "Hello Linux World" > mmapfile.txt
vaishali@LAPTOP-QQD0F0TI:~$ cat mmapfile.txt
Hello Linux World
vaishali@LAPTOP-QQD0F0TI:~$ nano mmap_program.c
vaishali@LAPTOP-QQD0F0TI:~$ gcc mmap_program.c -o mmap_program
/usr/bin/ld: /usr/lib/gcc/x86_64-linux-gnu/13/../../../x86_64-linux-gnu/Scrt1.o: in function `_start':
(.text+0x1b): undefined reference to `main'
collect2: error: ld returned 1 exit status
vaishali@LAPTOP-QQD0F0TI:~$ ./mmap_program
-bash: ./mmap_program: No such file or directory
```

```
  command 'car' from deb ucommon-utils (7.0.1-0.1)
  command 'phar' from deb php8.3-cli (8.3.6-0ubuntu0.24.04.9)
  command 'phar' from deb php-cli (2:8.2+93ubuntu1)
  command 'phar' from deb php8.2-cli (8.2.12-1ubuntu2)
  command 'pchar' from deb pchar (1.5-6)
  command 'shar' from deb sharutils (1:4.15.2-9)
See 'snap info <snapname>' for additional versions.
Command 'int' not found, but there are 29 similar ones.
-bash: syntax error near unexpected token `{'
-bash: !inSingle: event not found
> ^C
```

```
SRK:~$ ./shell
echo 'hello world'

echo 'hello     world'

echo '$HOME'

echo "hello world"

echo "$HOME"

echo ''

echo ""

echo 'hello world'

echo "hello world"
```

```
                buffer[j] = '\0';
                args[arg++] = strdup(buffer);
 }   return 0;ree(args[i]);gs[i] != NULL; i++) {== NULL) { && k <
 -bash: syntax error near unexpected token `('
 ^CCommand 'int' not found, but there are 29 similar ones.
 Command 'int' not found, but there are 29 similar ones.
 Command 'int' not found, but there are 29 similar ones.
 Command 'int' not found, but there are 29 similar ones.
 Command 'int' not found, but there are 29 similar ones.
 Command 'char' not found, did you mean:
   command 'chr' from snap chr (0.80.4)
   command 'chat' from deb ppp (2.4.9-1+1.1ubuntu1)
   command 'car' from deb ucommon-utils (7.0.1-0.1)
   command 'phar' from deb php8.3-cli (8.3.6-0ubuntu0.24.04.9)
   command 'phar' from deb php-cli (2:8.2+93ubuntu1)
   command 'phar' from deb php8.2-cli (8.2.12-1ubuntu2)
   command 'pchar' from deb pchar (1.5-6)
   command 'shar' from deb sharutils (1:4.15.2-9)
 See 'snap info <snapname>' for additional versions.
 Command 'int' not found, but there are 29 similar ones.
 -bash: syntax error near unexpected token `{'
 -bash: !inSingle: event not found
 > ^C
```